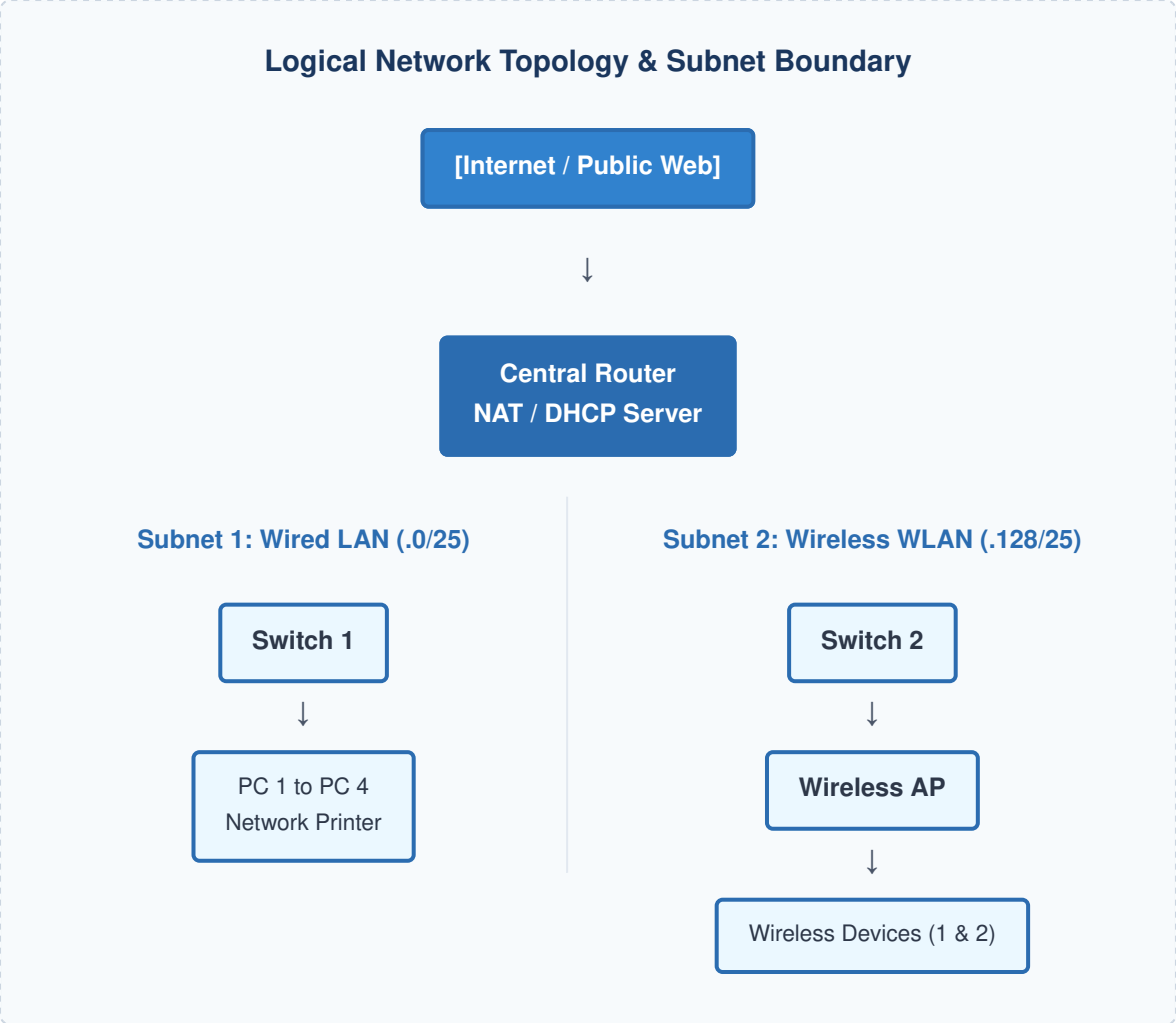
Network Architecture Report: LAN Design & Packet Forwarding

🌐 Executive Summary & System Overview

This technical report details the local area network (LAN) deployment for a single office floor. The architecture balances security, structural isolation, and scalability by partitioning a single physical environment into two distinct logical subnets using a Variable Length Subnet Mask (VLSM).

- **Primary Class C Private IP Range:** `192.168.10.0/24` (Subnet Mask: `255.255.255.0`)
- **Subnetting Strategy:** To separate trusted physical workstations from dynamic wireless devices, the primary block is divided into two distinct subnets using a `/25` mask (`255.255.255.128`):
  - **Subnet 1 (Wired LAN):** `192.168.10.0/25` (Supports host range .1 to .126)
  - **Subnet 2 (Wireless WLAN):** `192.168.10.128/25` (Supports host range . 129 to .254)



🛠️ Part 1: MAC & Frame Handling

1. MAC Address Assignment Plan

MAC addresses are hardcoded 48-bit physical identifiers represented in hexadecimal format.

Device	Interface	MAC Address
Router	LAN Interface 1 (Wired Gateway)	`00:1A:2B:3C:4D:01`
Router	LAN Interface 2 (Wireless Gateway)	`00:1A:2B:3C:4D:02`
PC 1	Internal Ethernet NIC	`A1:B2:C3:D4:E5:01`
PC 2	Internal Ethernet NIC	`A1:B2:C3:D4:E5:02`
PC 3	Internal Ethernet NIC	`A1:B2:C3:D4:E5:03`
PC 4	Internal Ethernet NIC	`A1:B2:C3:D4:E5:04`
Printer	Network Ethernet Interface	`B2:C3:D4:E5:F6:01`
Wireless Access Point (WAP)	Wired Uplink / Radio Interface	`C3:D4:E5:F6:A7:01`
Wireless Device 1	Wi-Fi Network Adapter	`D4:E5:F6:A7:B8:01`
Wireless Device 2	Wi-Fi Network Adapter	`D4:E5:F6:A7:B8:02`

2. Ethernet II Frame Structure (PC 1 to Printer)

When data travels across the Data Link Layer (Layer 2) from PC 1 to the printer on the same wired network, it is packaged into a standard Ethernet II Frame:

- **Preamble & SFD (8 bytes):** Pattern of alternating bits utilized exclusively for receiver clock synchronization.
- **Destination MAC Address (6 bytes):** `B2:C3:D4:E5:F6:01` (Directs the frame to the printer).
- **Source MAC Address (6 bytes):** `A1:B2:C3:D4:E5:01` (Identifies the originating sender, PC 1).
- **EtherType (2 bytes):** `0x0800` (Signals to the receiving hardware that the nested payload is an IPv4 packet).
- **Payload (46 to 1500 bytes):** The encapsulated Layer 3 data network packet containing the actual print command data.

- **Frame Check Sequence / CRC (4 bytes):** A cyclic redundancy check code computed by the source network card to verify data integrity upon arrival.

3. Collision Domain Management & Media Access Control

- **Wired Ethernet Environment:** Handled traditionally by CSMA/CD (Carrier Sense Multiple Access with Collision Detection). However, because dedicated modern network switches are used operating in Full-Duplex mode, every link forms an isolated collision domain. Collisions are 100% eliminated since bidirectional data transmission occurs simultaneously over discrete physical wire pairs.
- **Wireless Wi-Fi Environment:** Handled strictly via CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance). Because the airwaves are a shared medium operating in Half-Duplex, devices must listen to ensure the channel is clear. The device waits a random duration (Exponential Backoff) prior to broadcasting. To prevent collisions caused by the "Hidden Node Problem," devices use an RTS/CTS (Request to Send / Clear to Send) handshake to reserve the medium before sending heavy payloads.

🔢 Part 2: IP Addressing & Subnetting

1. IP Address Allocations

🔌 Subnet 1: Wired Infrastructure (192.168.10.0/25)

Subnet Mask: `255.255.255.128` | Default Gateway IP: `192.168.10.1`

Device Name	Assigned IP Address	Default Gateway
Router Interface 1	`192.168.10.1`	N/A
PC 1	`192.168.10.10`	192.168.10.1
PC 2	`192.168.10.11`	192.168.10.1
PC 3	`192.168.10.12`	192.168.10.1
PC 4	`192.168.10.13`	192.168.10.1
Printer	`192.168.10.20` (Static)	192.168.10.1

📶 Subnet 2: Wireless Infrastructure (192.168.10.128/25)

Subnet Mask: `255.255.255.128` | Default Gateway IP: `192.168.10.129`

Device Name	Assigned IP Address	Default Gateway
Router Interface 2	`192.168.10.129`	N/A
Wireless Access Point	`192.168.10.130` (Management)	192.168.10.129
Wireless Device 1	`192.168.10.140`	192.168.10.129
Wireless Device 2	`192.168.10.141`	192.168.10.129

2. The Functional Role of DHCP

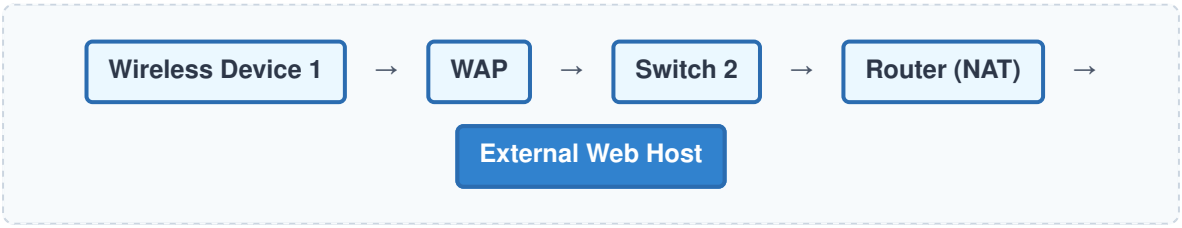
The DHCP (Dynamic Host Configuration Protocol) server, running on the central router, automates host deployment. Instead of configuring every network device manually, the DHCP server dynamically leases configuration blocks to clients instantly upon attachment. It hands out an available IP address, the local mask, the default gateway, and public DNS server addresses.

Strategic Exception: The network printer is intentionally isolated from the dynamic pool and explicitly assigned a Statically Configured IP. This locks its position on the network, guaranteeing that users can consistently reach the printer at a predictable network location.

🧭 Part 3: Routing & Packet Forwarding

Traffic Scenario: Wireless Device 1 (`192.168.10.140`) sends a request to an external web host (`8.8.8.8`).

1. End-to-End Hop Lifecycle



- **Step 1: Local Network Evaluation**
  Wireless Device 1 runs a logical AND mathematical operation on the destination IP (`8.8.8.8`) using its local subnet mask. It recognizes that the target host resides outside its local subnet. It determines it must forward this traffic block to its configured local Default Gateway (`192.168.10.129`). The device frames the IP packet into a wireless medium format, setting the target hardware destination field to the MAC address of the local Router interface (`00:1A:2B:3C:4D:02`).
- **Step 2: Layer 2 Intermediate Relaying**
  The Wireless Access Point (WAP) captures the radio frequency transmission, strips the 802.11 wireless headers, converts the core payload into a standard 802.3 Ethernet frame, and pushes it to Switch 2. Switch 2 reads the destination MAC address field, queries its local MAC Address Table, and bridges the frame directly to the dedicated physical port linked to the Router.
- **Step 3: Layer 3 Header Processing at the Router**
  The Router strips away the Layer 2 framing components and reads the inner IPv4 Network Header. It checks the Source IP (`192.168.10.140`) and Destination IP (`8.8.8.8`). The Router drops the integer value of the Time to Live (TTL) parameter by 1 to prevent packets from traveling in infinite routing loops across the public web.
- **Step 4: Next Hop Determination**
  The Router scans its internal Routing Table to find a match for the destination network (`8.8.8.8`). Finding no exact matching entry, the Router falls back onto its default catch-all route, the Gateway of Last Resort (`0.0.0.0/0`). This route forwards the packet out of its WAN interface toward the internet service provider's upstream node.
- **Step 5: Network Address Translation (NAT) Execution**
  The packet's private source address (`192.168.10.140`) cannot travel across the public internet because private IP ranges are non-routable. The Router uses NAT/PAT (Port Address Translation) to overwrite the internal private IP address with its own Public WAN IP Address assigned by the ISP. It also modifies the source port

number to map the connection and logs it in its internal NAT Translation Table. When the external web host responds, the Router looks at this table to translate the public address back into the private address of Wireless Device 1.